Aanidhya Patidar

+91 80858 58001 aanidhyapatidar@gmail.com
linkedin.com/in/notaanidhya github.com/notaanidhya

Summary

Computer Science (AI & ML) student who builds end-to-end systems — fine-tuned PyTorch classifiers, deployed FastAPI backends, and production PWAs with real users. Shipped two projects in agritech and one in deepfake detection, working across full-stack development, applied machine learning, and real-time systems architecture.

Education

Vellore Institute of Technology — *Bhopal, MP*
Bachelor of Technology in Computer Science (AI & ML) — *2023 – 2027*
GPA: **8.71**/10.0

Technical Skills

Languages: Python, C++, JavaScript, SQL

Frameworks & Libraries: FastAPI, Flask, React, Vite, PyTorch, Pandas, NumPy, Scikit-learn, LightGBM, SQLAlchemy, TanStack Query, i18next

Databases: PostgreSQL, SQLite, MySQL

Developer Tools: Git, GitHub, VS Code, Postman, Alembic, Docker, Vercel, Render, WebSockets, REST APIs

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Soft Skills: Team Leadership, Cross-Functional Collaboration, Technical Communication

Projects

Krishi Khata — Digital Farm Ledger & Market Intelligence | *FastAPI, React, Vite, PostgreSQL, Gemini AI* — *2025 – 2026*

- Built and deployed a mobile-first app for Indian farmers — digital accounting ledgers, real-time mandi price tracking, and AI crop advisory, with frontend on Vercel and backend on Render.
- Designed a JIT mandi price pipeline using government APIs, caching 30-day commodity histories per district in PostgreSQL for sub-200ms lookups.
- Integrated a bilingual AI engine via Google Gemini that intercepts Accept-Language headers and rewrites system instructions at runtime for native Hindi or English responses.
- Set up a hardened WebSocket gateway for real-time community chat with JWT-authenticated handshakes and SlowAPI rate limiting on write-heavy routes.
- Localized the React frontend across 70+ UI strings via i18next and built a Quick-Select Crop Hub that auto-fills search parameters from the farmer's active crops.

DeepGuard — Deepfake Face Detection System | *Python, PyTorch, EfficientNet-B4, OpenCV, MTCNN* — *2026*

- Fine-tuned EfficientNet-B4 on FaceForensics++ to classify real vs. AI-manipulated faces across face-swap and reenactment forgeries at multiple compression levels.
- Built an MTCNN face extraction pipeline in OpenCV to crop and align face regions before classification, reducing background noise and sharpening model focus.
- Applied Grad-CAM to surface which facial regions drove predictions, making outputs interpretable beyond a raw confidence score.
- Handled class imbalance via weighted cross-entropy loss and stratified splits across forgery subtypes; evaluated on held-out data using AUC-ROC alongside accuracy.

AgroSense v4 — Crop Yield Prediction Platform | *Python, LightGBM, Flask, React, Open-Meteo API* — *2025*

- ML & Backend Lead on a 6-person team — sole owner of model training, feature engineering, and Flask REST API design; coordinated cross-functional deliverables across frontend and data sourcing sub-teams.
- Built a LightGBM regression model on historical crop, soil, and rainfall data across multiple crops and districts; evaluated using MAE and R^2 per crop category rather than in aggregate.

Activities

Google Developer Groups on Campus — VIT Bhopal — *2024 – Present*

LeetCode: Solved 200+ problems across arrays, graphs, dynamic programming, and trees — leetcode

Certifications

Applied Machine Learning in Python — University of Michigan (Coursera) — *2024*

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — *2025*

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — *2025*